COVID-19 Global Impact Dashboard

Executive Summary | Data Analytics Project

Purpose

This project explores public COVID-19 time-series data to compare reported cases, deaths, and vaccination progress across countries and dates. It demonstrates a reproducible analytics workflow from data preparation to interactive visual communication.

Approach

Python and Pandas are used to parse dates, validate numeric columns, document missing values, handle negative daily revisions for trend displays, and derive cases/deaths per million, vaccination coverage, and rolling averages. A Streamlit dashboard provides country and date filters, trend charts, country comparisons, and CSV export.

Dashboard questions

- How do case and death trends change over time across selected countries?
- How do population-normalized rates change the interpretation of country comparisons?
- How does reported full-vaccination coverage vary by country?
- Does vaccination coverage appear associated with later mortality patterns? This is descriptive, not causal.

Data quality and assumptions

Invalid dates and missing location/date records are removed from the analytical table. Numeric conversion failures remain missing rather than being silently changed to zero. Negative daily revisions are clipped only for daily trend analysis. Rates use the population value provided by the source.

Data status and limitations

Important: The ZIP includes an illustrative dataset only so the dashboard can run offline. It is not real COVID-19 evidence. Run *python download_data.py* to retrieve the public Our World in Data dataset before reporting factual country findings. Cross-country reporting differences and confounding factors limit causal conclusions.

Expected deliverables

Cleaned dataset and documented assumptions; 3–5 findings written in plain language after running on public data; interactive dashboard; and this one-page executive summary.